

IMA Intelligence

Biomechanical day review · 8-band stride distribution · 3-tier CoD asymmetry

May 2026 · v1.0

Quick summary

IMA Intelligence answers a question the other monitoring pages don't: **"what did today's training actually deliver, biomechanically?"** Where Load Intelligence shows volume and Decel Intelligence flags chronic risk, IMA Intelligence describes the *shape* of a single session — how many strides at each velocity band, how the team's Change-of-Direction load split left-vs-right, which players got the heaviest sprint exposure compared to their own 14-day average.

Built on the Catapult Vector S7+ pod's IMA Free Running V2 algorithm. Every detected footstrike is bucketed by IMU-derived velocity into one of 8 bands; MicroPulse groups them into Low (b1-3, walking → medium running), Mid (b4-6, fast running → HSR), and High (b7-8, sprint → max sprint). The page sums them per player + team and overlays a 3-tier Change-of-Direction asymmetry profile (Bishop 2020).

1. What you see in the system

Page location

Sidebar → **Load Monitoring** → **IMA Intelligence** (`/coach/ima-intelligence`). Hidden on Lite-tier Catapult plans that don't have IMA Free Running enabled — Lite teams see HSR Intelligence in the same slot as the equivalent injury-risk surface.

Three layers, top to bottom

1. Session shape card (top)

A coloured card with the session classification headline, four KPI tiles, the team-level Change-of-Direction profile, and the 8-band stride distribution stripe. Reading this card alone gives you 80% of what you need to know about the day.

The session classification badge is one of four labels:

- **High-cadence day** — > 30% of strides in bands 5-8. Sprint or COD-heavy session.
- **Mixed day** — typical MD-2/MD-3 distribution. Most days fall here.

- **Tactical day** — > 65% of strides in bands 1-3, < 15% in 5-8. Walk-through, set-piece work, or possession at controlled tempo.
- **Recovery or short session** — < 1,500 strides total per player. Used for MD-1 walk-throughs, MD+1 light days, or off-pitch sessions.

2. Change-of-Direction profile (team)

Three stacked L/R bars (Low / Medium / High intensity), each scaled relative to the highest-volume tier. Asymmetry % is shown on the right; only the **high-intensity tier** is colour-flagged (Bishop 2020 — see Section 3).

3. Per-player breakdown table

Sortable table with one row per player. Columns: Player, Total strides, b1-3, b4-6, b7-8, Sprint (b5-8), vs baseline, CoD L/R (high tier), CoD asym (high tier). Click the chevron at the start of any row to expand a 3-tier CoD breakdown for that player.

2. The 8 IMA Free Running bands

Catapult IMA Free Running V2 detects every footstrike via the on-board IMU (gyro + accelerometer, 100 Hz) and buckets each stride by IMU-derived velocity into one of 8 bands. Standard team configuration uses 3 km/h increments above 15 km/h:

Band	Velocity (km/h)	Movement type	Coaching interpretation
1	0 – 8	Walking	Idle, walk-back, brief stops
2	8 – 12	Light jogging	Transitions, warm-up pace
3	12 – 15	Medium running	Game-pace running, possession
4	15 – 18	Fast running	Pressing, transitions
5	18 – 21	High-speed running	HSR threshold (Buchheit 2014)
6	21 – 24	HSR / sprint approach	Accelerated running, breakaways
7	24 – 27	Sprint	Sustained sprint efforts
8	27 +	Max-velocity sprint	Top speed, breakaway sprints

How MicroPulse groups the bands

- **Low (b1-3)** — strides at < 15 km/h. Dominates most training days. Not injury-relevant directly but reflects total movement volume.
- **Mid (b4-6)** — strides at 15-24 km/h. The HSR threshold lives inside this group (band 5 ≥ 18 km/h). Together with high bands forms the Sprint Exposure unit.
- **High (b7-8)** — strides at 24+ km/h. Pure sprint zone. Most hamstring strains in football happen during these strides (Duhig 2016).

- **Sprint (b5-8)** — the analytical unit used in Sprint Exposure metric (Malone 2018). All strides at 18+ km/h, regardless of whether the player sustained sprint speed or accelerated through it briefly.

3. Change-of-Direction asymmetry (Bishop 2020)

Catapult IMA detects Change-of-Direction events independently of stride bands, then classifies them by intensity (low / medium / high) and side (left / right). MicroPulse surfaces all 6 cells per player per day so coaches can read **directional load** separately from raw stride volume.

Why 3 tiers, not one

Bishop and colleagues (J Athl Train 2020) showed that L/R asymmetry > 15% on **high-intensity** CoD events is the strongest single predictor of non-contact lower-limb injury in soccer — roughly 3× injury risk multiplier. But asymmetry at lower intensities behaves differently:

Tier	Typical movement	Asymmetry meaning	Coach action threshold
Low	Walking turns, postural shifts	Often positional habit — preferred dominant side. Not injury-relevant.	Information only — not flagged.
Medium	Game-pace turns and cuts	Mix of habit + capability. Sustained asymmetry > 15% over weeks is worth tracking.	Information only at the day level — review chronic trends in Decel Intelligence.
High	Sharp cuts, full-speed direction changes	Reveals limb-specific capacity gaps. Bishop 2020: > 15% = 3× injury risk.	Flagged with colour. > 15% → CoD-specific work (single-leg eccentric, Nordic, Copenhagen). Cross-check VALD bilateral deficit.

Reading the team CoD profile (top card)

The team-level profile sums every player's CoD efforts across all tiers and shows three stacked bars:

- **Bar length** scales with absolute volume — the highest-volume tier sets full width, others scale down proportionally.
- **Left / right split** is shown via the blue (L) and purple (R) fill ratio.
- **Asymmetry %** shown to the right. Only the high tier is colour-flagged (rose > 15%, amber > 10%).

Low and medium tiers will *usually* show asymmetry < 10% (positional habit averages out across the squad). If you see medium-tier asymmetry > 15% consistently across multiple days, that's worth a conversation.

Reading per-player CoD breakdown (expandable rows)

Click the chevron (►) at the start of any player row to expand a 3-tier breakdown of their CoD profile for that day. Same bar logic as the team profile but scoped to one player. The key reading:

- **All 3 tiers asymmetric** — likely a consistent limb-dominance issue. Needs CoD-specific intervention.

- **Only high tier asymmetric** — capability gap revealed at speed. Strength + neuromuscular work.
- **Only low tier asymmetric** — positional habit (probably not the dominant foot on the ball). Not actionable on its own.

4. Per-player table columns

Column	What it means
Player	Click name to open Decel Intelligence drilldown for that player (chronic L/R asymmetry, Sprint Speed Drop, Sprint Exposure).
Total	Sum of all 8 bands. Reflects total stride volume. Used in session classification (< 1,500 = recovery day).
b1-3 / b4-6 / b7-8	Tier subtotals. Compare to the team's session classification — a player with very different distribution from the team mean may have been doing different work (e.g. rehab on the side).
Sprint (b5-8)	The Sprint Exposure unit. Sortable. Highest values are typically wingers and attacking midfielders.
vs baseline	Today's sprint-strides ÷ player's own 14-day training-only average × 100. Colour-coded: > 150% red (heavy day), 130-150% amber, 70-130% slate (normal), < 70% blue (light). An asterisk (*) means the baseline is built on only 1 training day — common right now because pre-Catapult-fix training days have no IMA data; resolves automatically as more days accumulate.
CoD L/R	High-tier CoD efforts only, left / right counts. Lower tiers visible in the expanded row.
CoD asym.	High-tier asymmetry %. Bishop 2020 threshold colour-coded: > 15% rose-bold, 10-15% amber, ≤ 10% emerald. Only computed when total high CoD ≥ 10 efforts (avoids noise on rest days).

5. When to open IMA Intelligence

After every training session (30 min after Catapult sync)

This is the page's primary use case. You designed the session to be MD-3 high-cadence work. Did it actually deliver? Open IMA Intelligence:

- **Check session classification badge** — does it match what you planned?
- **Scan the band stripe** — was the cadence distribution where you wanted it?
- **Look at the player table** — sort by Sprint (b5-8) and check whether your high-cadence intent reached the players you targeted, or whether substitutes ended up doing it.
- **Check High-CoD flagged KPI** — if 3+ players are flagged with > 15% high-tier asymmetry, you may have over-loaded one direction (think rondos rotating one way).

Before designing the next session

If yesterday's session was higher-cadence than planned, today's session should rebalance. Open IMA Intelligence for yesterday and check:

- **Players above 150% vs baseline** — these need a lighter day today.
- **Players with high-tier CoD asymmetry > 15%** — avoid loading the same direction today.

During MD-3 / MD-2 prep weeks

Compare consecutive days to see if you're hitting the periodization model:

- **MD-3 should be high-cadence day** (sprint + COD)
- **MD-2 mixed** (tactical + light power)
- **MD-1 tactical or recovery** (low cadence dominant)

If the classifications don't match the periodization plan, the players' actual load won't match what you intended.

After matches

Match-day classification is almost always high-cadence. The interesting comparison is across positions: how did your starting XI cluster compared to the bench? Open IMA Intelligence for the match date and sort by Sprint (b5-8) — if a starter is well below his usual match output, he likely needs a softer MD+1 / MD+2.

Indoor training

This page is one of the few that **works fully indoors**. IMA Free Running uses IMU data only — no GPS signal needed. When you train indoors and Indoor Load Intelligence / Decel Intelligence show degraded confidence, IMA Intelligence still has the full picture.

6. Practical example — Breiðablik May 12 (Monday training)

Real session from the Breiðablik 1st team — Monday 12 May 2026, MD+1 day after a Friday match. Coach used a recovery-oriented session with possession work in tight spaces.

What the session shape card showed

- **Session classification: Recovery or short session** — team total < 1,500 strides per player average.
- **Total strides:** 13,144 across 18 players (~730 per player).
- **Sprint-strides (b5-8):** 3,189 (24% of total).
- **L/R CoD balance (high):** 49 L / 61 R — 20% asymmetry team-wide.
- **High-CoD asym. flagged:** 2 / 18 players above 15%.

Band distribution

- **Low (b1-3):** 69% — dominated by walking and light jogging (expected for MD+1).
- **Mid (b4-6):** 9% — minimal fast running.
- **High (b7-8):** 22% — surprisingly high for a recovery day. Cause: short rondos with explosive direction changes triggered band 7-8 strides.

Players flagged for high-tier CoD asymmetry

Two players exceeded the Bishop 2020 threshold of 15% on high-intensity CoD:

- **Anton Logi Lúðvíksson** — 5 L / 6 R, 17% asymmetry (just over threshold; positional habit, monitor).
- **Höskuldur Gunnlaugsson** — 4 L / 11 R, **64% asymmetry**. Strong limb dominance signal — needs CoD-specific work on the left side (single-leg eccentric, Copenhagen adductor).

vs baseline takeaways

Two stand-out values in the sprint-strides vs baseline column:

- **Aron Bjarnason** — 286 sprint-strides today, 136% of 14-day baseline. Did more high-cadence work than usual on a recovery day. Flag for MD+1 monitoring.
- **Óli Valur Ómarsson** — 19% of baseline. Took it very easy. If he played 90 min on Friday this is appropriate, but check his Sprint Speed Drop chip on the Decision Summary.

Action items the coach could take from this view

- Assign **Höskuldur** a left-side CoD-focused recovery block (Copenhagen adductor 2×6 each side, single-leg eccentric step-ups 2×8 each side).

- Check whether **Aron's** elevated sprint-stride volume was deliberate (extra running between drills) or accidental (rondo intensity drifted up). If accidental, design today's rondos with wider spacing to keep cadence down.
- Send a short message to **Óli Valur** asking how he feels — his 19% baseline suggests he barely participated, which combined with > 60 match minutes on Friday is appropriate but worth confirming.

7. Confidence, limitations, and edge cases

vs-baseline confidence flags

During the Catapult Free Running transition (started May 2026), most training days before the fix have IMA = 0. That means a player's 14-day baseline often contains only 1 qualifying training day right now. We flag this with an asterisk (*) and italic formatting in the vs-baseline column. The asterisk resolves automatically as more post-fix training days accumulate.

CoD asymmetry minimum threshold

Asymmetry % is only computed when the player has ≥ 10 CoD efforts in that tier on that day. Below that, the column shows "—". This avoids reading injury risk from sample sizes too small to be meaningful (e.g. 2 L / 0 R looks like 100% asymmetry but is meaningless).

Players with no data row

If a player didn't wear their pod that day (or wasn't on the pitch), their row shows all dashes. The chevron is disabled. They don't contribute to the team session card. This is correct behaviour — manual entries fill in distance/load but not granular IMA bands, so they can't be meaningfully classified.

Days before the Catapult Free Running fix

Activities ingested before Catapult enabled Free Running config in OpenField don't have IMA band data — bands 1-8 are all 0 even though GPS V5/V6 distance is populated. These days show 0 total strides on IMA Intelligence and don't contribute to the per-player baseline. Once a player has ≥ 1 post-fix training day with IMA data, their baseline starts working.

Session classification on small squads

With < 8 players captured, the team-level session classification is less stable (one player with very different work shifts the average). Per-player classification is still reliable — focus on the table rather than the team card on small-squad days.

8. Frequently Asked Questions

Why is IMA Intelligence in Load Monitoring, not Injury Monitoring?

IMA Intelligence describes the **shape** of a session — what kind of work was actually delivered. That's a load question, not an injury question. Injury Monitoring (Injury Pattern Analysis, RTP) is for tracking actual injuries and threshold alerts. The CoD asymmetry flag on IMA Intelligence is informational; chronic injury risk is summarised on Decel Intelligence.

How does this differ from Decel Intelligence?

Decel Intelligence is a **chronic injury-risk dashboard** — it averages CoD asymmetry over the last 14 days, surfaces the worst-trending players, and shows Sprint Speed Drop / Sprint Exposure together. IMA Intelligence is a **single-day biomechanical snapshot** — what happened today, who got the most volume, where the CoD load went left vs right. Use IMA Intelligence after each session, Decel Intelligence for weekly chronic-risk review.

Why does the band stripe say 22% high on a recovery day?

IMA estimates velocity per stride using IMU stride-rate $\times$ stride-length, not GPS. On a possession-heavy day with quick direction changes and explosive 2-3-stride bursts, the IMU can register many strides as high-velocity even when GPS shows minimal sustained sprint distance. This is actually a strength of IMA — it captures eccentric load from quick footwork that GPS misses (McBurnie 2022).

A player has IMA b7-8 = 0 but ran 200m on GPS at > 25 km/h. Why?

Almost always because the activity was ingested before Catapult enabled Free Running config in OpenField. IMA Free Running is a separate algorithm from GPS velocity bands — it needs the Free Running config enabled at the time of activity processing. GPS V5/V6 always work; IMA bands only fill in when Free Running was on. Going forward (post-fix) both populate.

How accurate is the per-player baseline?

With 2+ training days in the 14-day window: $\pm 20\%$ accuracy. With only 1 training day (asterisk flag): $\pm 50\%$ — interpret with caution. With 5+ training days: $\pm 10\%$. The system auto-strengthens the baseline as more post-fix training days come in.

Can I see historical days?

Yes — use the date picker in the top-right. Goes back as far as your team has Catapult data. Useful for after-match reviews (compare match vs typical training profile) or for spotting periodization drift over a week.

Why is the high-tier asymmetry threshold 15%?

Bishop et al. (J Athl Train 2020) followed 39 elite soccer players over 12 months and found > 15% asymmetry on high-intensity CoD tasks tripled non-contact lower-limb injury risk vs < 15%. The threshold is widely cited and used in published team-load monitoring systems. Lower thresholds (10%) catch more players but with more false positives; higher (20%) misses real risks. 15% is the consensus actionable cutoff.

9. References and further reading

Key research

Bishop C, Read P, Bromley T, et al. (2020). The association between inter-limb asymmetry and injury risk in soccer players. **J Athl Train** 55:1062–1070.

McBurnie AJ, Harper DJ, Jones PA, et al. (2022). Deceleration training in team sports. **Sports Med** 52:1473–1487.

Buchheit M, Allen A, Poon TK, et al. (2014). Integrating different tracking systems in football. **J Sports Sci** 32(20):1844–1857.

Buchheit M, Cardinale M. (2019). Adding heat to the live-load monitoring debate. **Sport Performance & Science Reports** #59.

Duhig S, Shield AJ, Opar D, et al. (2016). Effect of high-speed running on hamstring strain injury risk. **Br J Sports Med** 50(24):1536–1540.

Malone S, Hughes B, Doran DA, et al. (2018). Can the workload–injury relationship be moderated by improved strength, speed and repeated-sprint qualities? **J Sci Med Sport** 21(7):710–715.

Technical references

Catapult Sports (2024). Vector S7+ IMA Free Running V2 — Algorithm Documentation. Catapult Cloud documentation.

Related MicroPulse guides

- MicroPulse Coach Guide: Sprint Exposure (IMA) — companion volume-side metric on bands 5-8
- MicroPulse Coach Guide: Decel Intelligence (McBurnie 2022) — chronic 14-day injury-risk dashboard
- MicroPulse Coach Guide: Indoor Load Intelligence (FMP / Brown 2016) — load-side counterpart
- MicroPulse Catapult-IMA Briefing — technical deep-dive on the IMA Free Running V2 algorithm